

In the first phase of a study of the cost of transporting milk from farms to dairy plants, a survey was taken of firms engaged in milk transportations. Cost data on X_1 = fuel, X_2 = repair and X_3 = capital, all measured on a per mile basis, are presented in below table for $n_1=20$ gasoline and $n_2=15$ diesel trucks.

Table-1: Milk Transportation Cost Data

Gasoline Trucks			Diesel Trucks		
X_1	X_2	X_3	X_1	X_2	X_3
16	12	11	8	12	9
7	3	4	7	5	17
10	2	10	10	3	11
4	6	8	10	15	6
11	5	11	13	4	29
14	6	10	10	13	11
13	11	11	6	9	19
13	14	9	11	10	14
29	15	3	9	3	14
13	8	10	10	5	21
7	6	8	11	18	35
10	4	9	12	12	17
10	5	10	9	13	21
11	6	8	8	10	17
12	14	14	8	6	16
10	3	6			
10	6	12			
9	3	12			
12	8	11			
8	14	12			

Test whether mean cost vectors for Gasoline Trucks and Diesel Trucks are same or not at $\alpha = 0.01$ level of significance

$$[F_{3,31}(0.01) = 4.48].$$

Step 1 :- Hypothesis Statement :

Null Hypothesis : The mean cost of Gasoline Trucks and Diesel Trucks are same .

$$H_0 : \mu_G = \mu_D$$

Alternate Hypothesis : The mean cost of Gasoline Trucks and Diesel Trucks are significantly difference .

$$H_1 : \mu_G \neq \mu_D$$

Step 2 :- Data Analysis :

```
library(readxl)

data<-read_excel("D:/MULTIVARIATE WORK/MUV-7.xlsx")

## New names:
## • `X1` -> `X1...1`
## • `X2` -> `X2...2`
## • `X3` -> `X3...3`
## • `X1` -> `X1...4`
## • `X2` -> `X2...5`
## • `X3` -> `X3...6`
```

Gasoline Trucks

```
g_x1<-data$X1...1
```

```
g_x2<-data$X2...2
```

```
g_x3<-data$X3...3
```

Diesel Trucks

```
d_x1<-c(8,7,10,10,13,10,6,11,9,10,11,12,9,8,8)
d_x2<-c(12,5,3,15,4,13,9,10,3,5,18,12,13,10,6)
d_x3<-c(9,17,11,6,29,11,19,14,14,21,35,17,21,17,16)
```

Calculating gasoline r1

```
mean_d1<-mean(d_x1)
```

```
mean_d2<-mean(d_x2)
```

```
mean_d3<-mean(d_x3)
```

```
d1_matrix<-matrix(data = c((d_x1-mean_d1),(d_x2 - mean_d2),(d_x3 - mean_d3)),ncol=3,byrow = FALSE)
```

```
r1<-t(d1_matrix)%*(d1_matrix)
r1
```

```
##           [,1] [,2]      [,3]
## [1,] 49.73333 14.6  61.06667
## [2,] 14.60000 306.4  41.60000
## [3,] 61.06667 41.6  779.73333
```

Calculating gasoline r2

```
mean_g1<-mean(g_x1)
```

```
mean_g2<-mean(g_x2)
```

```
mean_g3<-mean(g_x3)
```

```
g1_matrix<-matrix(data = c((g_x1-mean_g1),(g_x2 - mean_g2),(g_x3 - mean_g3)),ncol=3,byrow = FALSE)
```

```
r2<-t(g1_matrix)%*(g1_matrix)
r2
```

```
##           [,1]      [,2]      [,3]
## [1,] 466.95 222.05 -72.05
## [2,] 222.05 342.95  32.05
## [3,] -72.05  32.05 140.95
```

Calculating Inverse

```
s=(r1+r2)/(20+15-2)
```

```
inverse_s=solve(s)
```

Calculating t-square

```
x1_matrix<-matrix(data=c(g_x1,g_x2,g_x3),ncol=3,byrow=FALSE)
x1<-colMeans(x1_matrix)
```

```
x2_matrix<-matrix(data=c(d_x1,d_x2,d_x3),ncol=3,byrow=FALSE)
x2<-colMeans(x2_matrix)
```

```
t_sq<-((20*15)%*%t(x1-x2)%*%inverse_s)%*%(x1-x2))/35
t_sq
```

```
##           [,1]
## [1,] 22.01801
```

F_calculated

```
f_cal<-((t_sq)*(35-3-1))/((35-2)*3)
f_cal
```

```
##           [,1]
## [1,] 6.894529
```

Step 3 :- Conclusion :

So, at the $\alpha = 1\%$ level of Significance

$F_{cal} = 6.89 > 4.48$, by according to the given data we rejected the Null Hypothesis .

Step 4 :- Interpretation :

The mean cost of Gasoline Trucks and Diesel Trucks are significantly differ .